(B) Define Device driver. What are the functions of device driver? Briefly explain the interface between device independent part of operating system and device driver [03] 6U

OR

(B) What do you mean by a file system consistency? How Unix kernel maintain file system consistency? [03] 5U

(C) Briefly explain protection mechanism used in operating systems. Also discuss Access control list and Capabilities [02] 5,6U

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ID No._____

Birla Vishvakarma Mahavidyalaya (Engineering College)

(An Autonomous Institution)

Third Year, B.Tech. (Computer Engineering)

End Semester Examination, Winter - 2022 (AY 2022-23)

Course Code: 3CP04 **Course Title: Operating Systems**

Date: 14.12.2022 **Time: 11:00 am to 1:30 pm** **Maximum Marks: 60**

Instructions:

- Each section must be written in a separate answer book.
- Numbers written in the square brackets to the right indicate maximum marks.
- Make assumptions wherever required
- The text just below marks indicates the Course Outcome Nos. (CO) followed by the Bloom's taxonomy level of the question, i.e., R: Remember, U: Understand, A: Apply, N: Analyze, E: Evaluate, C: Create

Section – I

Q. 1 (A) Compare monolithic, layered and micro kernel operating system approaches with their advantages and disadvantages [06] 1U

(B) What is thread? What are the differences between thread and process? Briefly explain three implementation of thread with their advantages and disadvantages [05] 2U

Q 2 (A) What is Monitor? Why is it known as higher level synchronization? Briefly discuss the mutual exclusion and synchronization in producer-consumer problem using monitor. Write complete pseudo code [06] 2U

OR

(A) What are the advantages of message passing mechanism over shared memory inter-process communication? Briefly discuss design issues of message passing mechanism. Consider two concurrently running processes: P1 with a statement S1 and P2 with a statement S2. Suppose we require that S2 be executed only after S1 has completed. Achieve above synchronization using message passing mechanism. Write complete pseudo code [06] 2U

(B) Briefly explain **negating** Mutual exclusion and Circular wait condition to prevent dead lock [03] 2U

(C) Briefly explain memory management with swapping. How does operating system keep track of memory usage? Briefly explain [04] 4U

OR

(C) Consider the following heap (Figure 1) in which blank regions are not in use and hatched region are in use. Apply First fit, Next fir, Best fit and Worst fit for flowing requests [04] 4A

300, 25,125 and 50

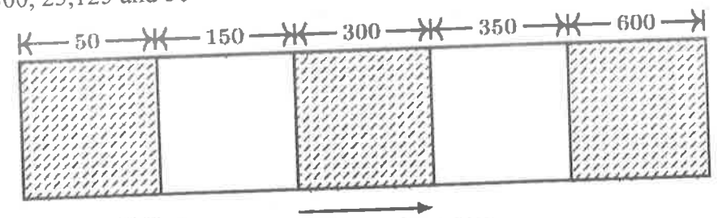

Figure 1

Q 3 (A) Consider a following process scenario in which each process first executes on CPU for given time duration then goes for I/O operation and then again executes on CPU before termination. CPU uses preemptive SRTF (Shortest Remaining time first) algorithm. Consider each process has its separate set of I/O devices to work in parallel with other process [04] 3A

Process	Arrival Time	CPU Burst Time	I/O Time	CPU Burst Time
P1	0	3	2	2
P2	0	1	3	1
P3	3	3	1	2
P4	6	5	4	5

Calculate average Turn-around time and Average waiting Time. Draw Gantt chart

(B) Consider the following state of a system with Five processes P0, P1, P2, P3, and P4, and four types of resources, A, B, C and D [05] 2A

Allocation

	A	B	C	D
P0	0	0	1	2
P1	1	0	0	0
P2	1	3	5	4
P3	0	6	3	2
P4	0	0	1	4

Maximum

	A	B	C	D
P0	0	0	1	2
P1	1	7	5	0
P2	2	3	5	6
P3	0	6	5	2
P4	0	6	5	6

Available

A	B	C	D
1	5	2	0

Define safe and unsafe state. Apply banker algorithm and verify that given state is safe or not? Show all calculations and show sequence of process. If a request from process P1 arrives for (0, 4, 2, 0) can the request be granted immediately? Show all calculations and sequence of process.

(C) What is starvation? How starvation problem is solved in priority based scheduling algorithm? [02] 3U

Section – II

Q. 4 (A) What are the advantages and disadvantages of memory management with Virtual memory? How virtual address is converted to physical address? Briefly explain the solutions used to make virtual to physical address mapping fast and to reduce the memory required for page table [06] 4U

(B) Assume that there are 4 page frames which are initially empty. Consider the reference string is 5 0 2 1 0 3 0 2 4 3 0 3 2 1 3 0 2. Find number of page faults if following page replacement algorithms are used [06] 4A

1. Optimal
2. LRU
3. Second chance

Show the contain of page frame after every memory reference

Q 5 (A) Define working set. Briefly explain WSCLOCK (working set clock) page replacement algorithm [04] 4U

(B) Briefly explain the different ways of implementation of files and directories in operating systems [04] 5U

OR

Q 5 (A) Why Page size should not be too small not too large? Derive the formula of optimal page size [04] 4U

(B) Consider following files with their starting block and length. Show the allocation if continuous allocation is used. Disk size is 36 blocks. How many files having size of 2 blocks can be accommodated in above disk? [04] 5A

File name	Starting Block	Length
P1	2	3
P2	17	1
P3	6	8
P4	1	1
P5	15	2
P6	29	2
P7	20	6

Q 5 (C) Briefly explain the methods used by an operating system to keep track of free blocks. Suppose that disk addresses require D bits. For a disk with a total of B blocks, F of which are free, state the condition under which the free list uses less space than the bit map. For D equal to 16 bits, express your answer as the percentage of disk space that must be free so link list required less space. [04] 5A

Q 6 (A) Suppose a disk has 200 cylinders numbered from 0 to 199. At some time, disk arm is at cylinder 76 and there is a queue of disk access requests for cylinders: 54, 32, 108, 22, 67, 77, 88, 42, and 82. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling algorithms? Assume that SCAN (Elevator) algorithm moves towards 199 when it starts execution [06] 6A

(i) Shortest seek first (SSF) (ii) SCAN (Elevator) (iii) CSCAN (iv) LOOK